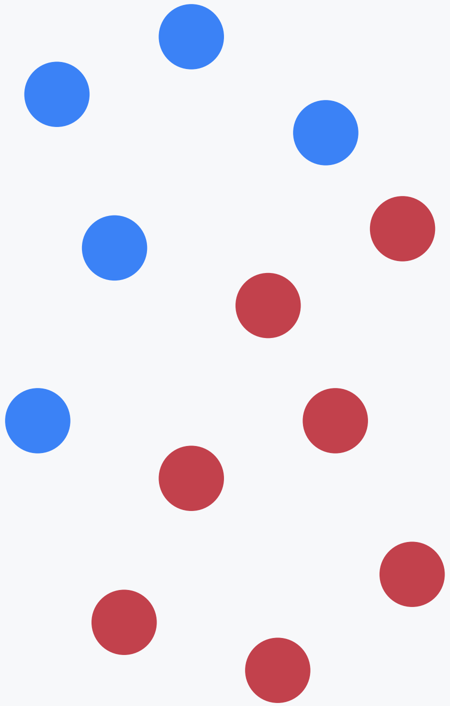
LAB MEETING / FIGURE-LED RESEARCH REPORT

单细胞测序中的治疗反应信号

From sample preparation to response modeling

Illustrative lab dataset
Smith et al., 2024 / protocol note

CELL-STATE SIGNAL MAP

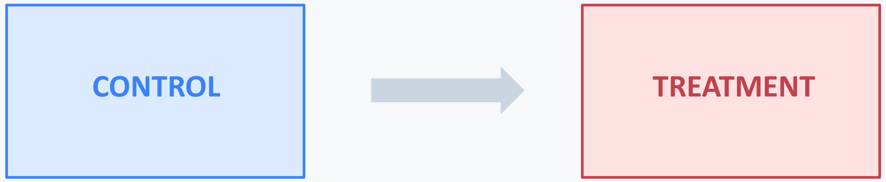


control → treatment response

01 / Research question

What changes when the treatment condition is introduced?

Can single-cell signals separate response states?



Hypothesis: treatment shifts the response index across cell populations.

same assay / different response state

SIGNAL	single-cell expression	READOUT	response index	TEST	replicate + validate

Figure question: the comparison is illustrative and does not establish clinical efficacy.

02 / Methods

Three stages connect the sample to an interpretable response index

A → B → C

A — SAMPLE PREPARATION

normalize input
and preserve cell state

B — SINGLE-CELL SEQUENCING

capture expression
across populations

C — RESPONSE MODELING

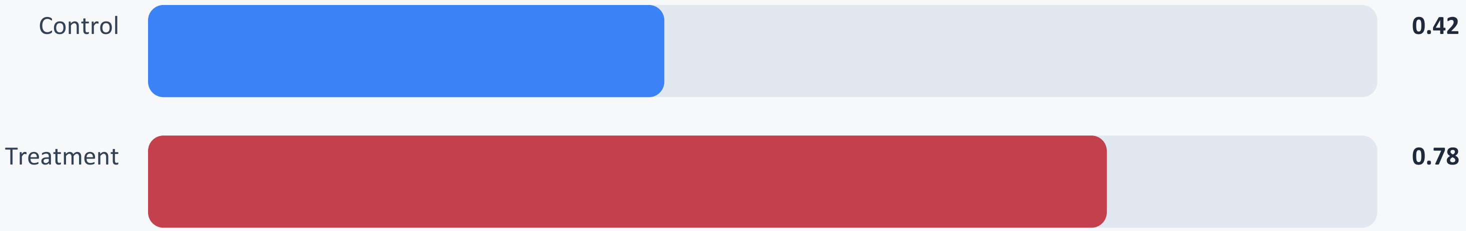
estimate response
with uncertainty

Methods overview. Stage labels are structural; protocol details remain in the source record.

03 / Results

Treatment response index is higher in the illustrative dataset

FIGURE 1



READOUT

0.78

treatment index

DIFFERENCE

0.36 index points

INTERPRETATION

direction matches hypothesis

BOUNDARY

illustrative; replicate

Figure 1. Illustrative control/treatment response index. Source: illustrative lab dataset; protocol adapted from Smith et al., 2024.

04 / Discussion

The result is promising, but interpretation remains bounded

WHAT THE FIGURE SUPPORTS

Treatment shows a higher response index.
The signal merits follow-up modeling.

Observed direction: 0.42 → 0.78

WHAT REMAINS UNCERTAIN

The dataset is illustrative, not a clinical conclusion.
No causal effect is established.

Evidence type: illustrative dataset

NEXT VALIDATION

Replicate across donors and batches.
Test robustness across cell states.

Validation unit: donor × batch

Discussion rule: separate an observed signal, a plausible interpretation, and a claim that still requires validation.

Discussion guardrail: distinguish an observed signal from a validated treatment effect.

05 / Next step

Preserve traceability before expanding the claim

NEXT EXPERIMENT

Replicate across donors and batches.

Source note

Illustrative lab dataset

Protocol adapted from Smith et al., 2024.

Figure 1 remains a signal, not a conclusion.

End of report / questions and discussion